

The empirical recurrence for OEIS A233945: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A233945 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A233945 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with every 2×2 subblock having the sum of the absolute values of the edge differences equal to 4.”. It has offset 1 and begins

5874, 180972, 5601514, 176523392, 5575336614, 176443282254, 5586328225626, 176913068237196, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:05:37, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	46	c_{19}	308168937770806549	c_{37}	−219379586347022152275471	c_{55}	1271312
c_2	71	c_{20}	2807261020208438482	c_{38}	−16501565103174519315503	c_{56}	−377574
c_3	−24582	c_{21}	−4575159457698867161	c_{39}	246002301127635217177226	c_{57}	−72725
c_4	149960	c_{22}	−25589254692966058150	c_{40}	−11669217086638390078783	c_{58}	214141
c_5	4969127	c_{23}	47798386896126291124	c_{41}	−208795863770623747222401	c_{59}	246574
c_6	−45376772	c_{24}	176119829500479566816	c_{42}	28709059910695233044975	c_{60}	−7399
c_7	−499219081	c_{25}	−366366861508713183700	c_{43}	133420202057032077602516	c_{61}	−4203
c_8	5827262754	c_{26}	−917062958130710650086	c_{44}	−26968755599467411264703	c_{62}	1421
c_9	27766523169	c_{27}	2103801084572335720089	c_{45}	−63752534294229344563021	c_{63}	1938
c_{10}	−424985142009	c_{28}	3603784810982217553428	c_{46}	15734205485054674985870	c_{64}	−128
c_{11}	−853573213006	c_{29}	−9153071401843702905308	c_{47}	22577225950613105295287	c_{65}	193
c_{12}	19784947271260	c_{30}	−10603041472431806804172	c_{48}	−6225747956774039497853	c_{66}	36
c_{13}	10472518453452	c_{31}	30348031734569022305711	c_{49}	−5853520199675875054646	c_{67}	−
c_{14}	−627259532230762	c_{32}	22973222430386283117250	c_{50}	1711546933218667196553	c_{68}	−
c_{15}	224899952924039	c_{33}	−76864189280904853575082	c_{51}	1092107158905642539101	c_{69}	−
c_{16}	14104215932435385	c_{34}	−35422793097519489022077	c_{52}	−327207130199022263337	c_{70}	−
c_{17}	−13182284663072271	c_{35}	148709046644792495449450	c_{53}	−143092152977765852028		
c_{18}	−231049112686798671	c_{36}	35711054045684697485869	c_{54}	42945128397044832179		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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